The American Journal of Chinese Medicine, Vol. 42, No. 5, 1099–1109

Institute for Advanced Research in Asian Science and Medicine
DOI: 10.1142/S0192415X14500682

Electro-Acupuncture Decreases Postoperative Pain and Improves Recovery in Patients Undergoing a Supratentorial Craniotomy

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Published 28 August 2014

Abstract: We performed this study to examine the effect of electro-acupuncture (EA) on postoperative pain, postoperative nausea and vomiting (PONV) and recovery in patients after a supratentorial tumor resection. Eighty-eight patients requiring a supratentorial tumor resection were anesthetized with sevoflurane and randomly allocated to a no treatment group (Group C) or an EA group (Group A). After anesthesia induction, the patients in Group A received EA at LI4 and SJ5, at BL63 and LR3 and at ST36 and GB40 on the same side as the craniotomy. The stimulation was continued until the end of the operation. Patient-controlled intravenous analgesia (PCIA) was used for the postoperative analgesia. The postoperative pain scores, PONV, the degree of dizziness and appetite were recorded. In the first 6 hours after the operation, the mean total bolus, the effective times of PCIA bolus administrations and the VAS scores were much lower in the EA group ($p < 0.05$). In the EA group, the incidence of PONV and degree of dizziness and feeling of fullness in the head within the first 24 hours after the operation was much lower than in the control group ($p < 0.05$). In the EA group, more patients had a better appetite than did the patients in group C (51.2% vs. 27.5%) ($p < 0.05$). The use of EA in neurosurgery patients improves the quality of postoperative analgesia, promotes appetite recovery and decreases some uncomfortable sensations, such as dizziness and feeling of fullness in the head.

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Keywords: Electro-Acupuncture; Supratentorial Craniotomy; Postoperative Pain; Appetite; Dizziness.

Introduction

In recent decades, more attention has been given to postoperative pain after neurosurgery. In particular, the supratentorial craniotomy has been studied because 80% of patients may experience mild to severe acute pain (Imaev *et al.*, 2013). In addition, increased postoperative pain is associated with hyper-intracranial pressure, postoperative bleeding, dizziness, anorexia and other symptoms (Yeh *et al.*, 2010; Bilotta *et al.*, 2013). Patient-controlled intravenous analgesia (PCIA) is always utilized to control postoperative pain in these patients and to provide a higher degree of satisfaction regarding pain relief (Angster and Hainsch-Müller, 2005; Pang *et al.*, 2013). Although PCIA has significant analgesic effects, postoperative nausea and vomiting (PONV), respiratory inhibition and somnolence, which are induced by opioids and may restrict the early neurological function exam, continue to be a problem. Therefore, it is important to provide neurosurgical patients with effective postoperative pain treatment while avoiding PONV and over sedation.

EA has been combined with general anesthesia to alleviate PONV (Grube *et al.*, 2009; Wang *et al.*, 2010; Lin and Chen, 2012), improve postoperative analgesia (Lin and Chen, 2009; Yeh *et al.*, 2010; Langenbach *et al.*, 2012) and reduce the intraoperative opioid requirement. The effect of acupuncture on the general anesthetic requirements has been previously studied in volunteers (Chernyak *et al.*, 2005; Kumnerddee and Pattapong, 2012) and during clinical anesthesia for different types of surgery (Liodden and Norheim, 2013). However, these studies rarely focused on postoperative pain treatment using acupuncture combined with traditional PCIA. In addition, postoperative pain after neurosurgery is always combined other symptoms, such as dizziness, feeling of fullness in the head, and these accompanying symptoms cannot be controlled by traditional analgesic methods such as PCIA. And postoperative pain also induces poor appetite, which was always ignored by the investigators.

The aim of this controlled, randomized, double-blind study was to examine the effect of EA combined with PCIA on postoperative pain after a supratentorial craniotomy. We hypothesized that EA would improve the quality of postoperative pain treatment. We also examined recovery variables such as dizziness, PONV and appetite as secondary outcome measures.

Materials and Methods

Patients

Following approval from the Human Research Committee of Beijing Tiantan Hospital, Capital Medical University and the receipt of informed consent, 88 ASA I–II adult patients aged 18–60 years who were undergoing supratentorial tumor resections under general anesthesia were enrolled in this study. All the subjects involved were blinded regarding the study, and the doctor who observed the postoperative pain and other outcomes was also blinded to the study.

Patients who were obese (BMI $> 30\,kg/m^2$), pregnant, taking any medication or had cardiovascular or respiratory diseases were excluded. Additionally, all the patients underwent a needling test before they were included in the study to exclude subjects who were resistant to acupuncture. Patients were also excluded if they developed a postoperative hematoma or infection. Patients were randomly allocated into 2 groups using a computer-generated randomized number table; Group C was a control group that did not receive EA during the operation ($n = 43$) and Group A was the experimental group that received EA ($n = 45$).

Acupuncture

We collaborated with an acupuncturist from Beijing University of Chinese Medicine to perform this study. In Group A, the acupuncture needles (0.25×40 mm) were inserted into acupoints at LI4 (Hegu), SJ5 (WaiGuan), BL63 (JinMen), LR3 (TaiChong), ST36 (ZuSanLi) and GB40 (QiuXu) on the same side as the craniotomy. For LI4, SJ5, BL63, LR3, ST36, and GB40, each acupuncture point was treated with one needle. LI4 and SJ5, BL63 and LR3, as well as ST36 and GB40, were connected in pairs to the acupuncture point nerve stimulator. EA stimulation was delivered via a LH202H HANS acupuncture point nerve stimulator (Beijing Huawei Co Ltd; Beijing, China) using a dense-dispersed wave with frequencies of 2 and 100 Hz alternating every 3 s. The waveform was a symmetric biphasic curve. The stimulation intensity was set to the level of maximum tolerance for each patient, and the stimulation lasted from the induction of anesthesia until the end of the operation. In Group C the acupuncturist did not insert any needles.

Anesthesia and Postoperative Analgesia

Following pre-oxygenation, anesthesia was induced in both groups with an IV infusion of sufentanil 0.3 μg/kg and propofol 2 mg/kg. A tracheal intubation was facilitated with IV vecuronium 0.1 mg/kg. General anesthesia was initially maintained with sevoflurane, 2% in oxygen at 2 L/min. At the end of the operation, sevoflurane was discontinued and the intravenous PCIA pump was connected. Patients were sent to the postoperative anesthesia care unit (PACU), where the PCIA device was explained to each patient, and its operation was reviewed prior to initiating the PCIA therapy. The PCIA device was programmed and contained 10 μg/kg fentanyl and 8 mg ondansetron with normal saline for a total of 100 mL without basal infusion. Supplemental bolus doses of 1 mL could be administered with a minimal lockout interval of 15 min and a maximum hourly dose of 4 mL if the patient was unable to achieve adequate pain relief from the PCIA device. At 48 hours, the PCIA therapy was discontinued in all the patients. All the patients started a liquid food diet in the morning after surgery.

Data Recording

All the patient and surgical variables were monitored and recorded. These included age, gender, weight, fluids administered, urine output, blood loss and the duration of the operation.

The postoperative pain score at rest was assessed with a standard 10 point visual analog scale (VAS, 0 = no pain to 10 = the worst pain imaginable). The incidence of PONV was recorded. Dizziness and feeling of fullness in the head were assessed at 6, 24 and 48 hours (no dizziness, mild dizziness, moderate dizziness, severe dizziness). Appetite was also assessed at 24 and 48 hours (1 = unable to eat, 2 = very few liquid foods or water, 3 = sufficient liquid foods or some solid foods, 4 = normal eating).

The total PCIA boluses, the VAS value, PONV and the need for rescue anti-emetic drugs were recorded at 1-, 2-, 3-, 4-, 6-, 24- and 48-hour intervals during the postoperative period. Appetite and dizziness were recorded at 24 and 48 hours.

Statistical Analysis

Based on our previous study, 30 patients were required in each group to achieve an 80% chance of detecting a significant difference in the VAS scores with a 5% level of significance (An *et al.*, 2010a,b; Yang *et al.*, 2012). The data were tested for normal distribution using the Kolmogorov–Smirnov test. The number of PCIA demands, the doses of rescue analgesic and the VAS values are presented as the mean $\pm$ SD, and differences were assessed using a one-way analysis of variance (ANOVA). A p value < 0.05 was considered statistically significant. The occurrence of PONV was assessed using the R $\times$ C χ^2 test. The ratio of dizziness and eating status were compared using the *Mann–Whitney U test*. The data entry and analysis were performed by statistical experts from Capital Medical University using the SPSS 19.0 statistical software.

Results

A total of 88 subjects participated in the study. Eighty-one subjects completed the study, and 7 were excluded (2 underwent surgery again, 2 remained unconsciousness after surgery, and 3 lacked complete information). The patient and surgical characteristics are shown in Table 1. There were no significant differences between the 2 groups regarding age, weight, gender or type and duration of surgery. Fluid administration, urine output and estimated blood loss were also similar between the 2 groups.

Effects of EA on Postoperative Pain

Figure 1 indicates that the pain level tended to decrease in both groups. However, during the first 6 hours, the VAS score of the EA group was much lower than that of the control group ($p < 0.05$). We then calculated the mean total PCIA bolus administration and the effective number of PCIA bolus administrations within the first 6 hours and over

Table 1. Patient and Surgical Characteristics

	Group A ($n = 41$) (Electro-Acupuncture Group)	Group C ($n = 40$) (Control Group)
Age (yr)	40.7 ± 12.1	39.1 ± 10.9
Body weight (Kg)	63.6 ± 9.5	65.5 ± 9.6
Male/female	17/24	21/19
Duration of surgery (min)	201.2 ± 38.6	210.5 ± 56.1
Duration of anesthesia (min)	262.3 ± 59.7	284.5 ± 107.2
Type of surgery		
Meningioma	11	17
Glioma	23	11
Metastasis or others	7	12
Dose of sufentanil (μg)	36.78 ± 1.33	38.12 ± 3.56
Fluid administration (ml)	3057.4 ± 574.2	3448.1 ± 513.1
Urine output (ml)	579.31 ± 363.1	584.6 ± 409.5
Estimated blood loss (ml)	1503.4 ± 611.4	1619.2 ± 813.8

Notes: Data are expressed as mean ± SD or numbers. No statistically significant differences were observed between the two groups.

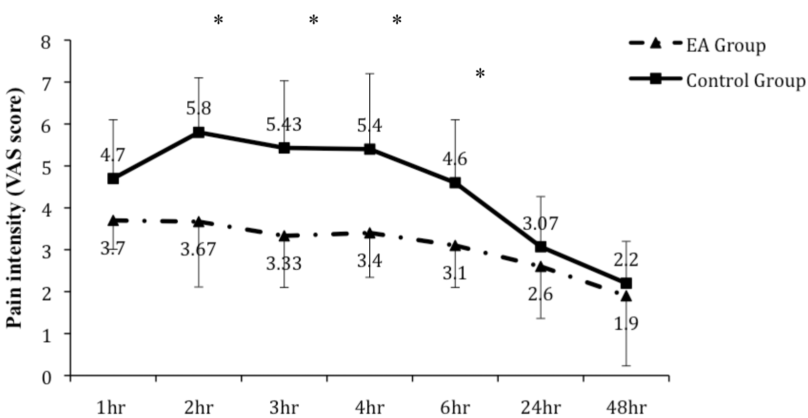

Figure 1. Changes in pain intensity (VAS values) over time. The data are presented as the mean visual analog score and the error bars represent SD. *$p < 0.05$ vs. the control group. EA group = Electro-acupuncture group.

6–48 hours. We found that the mean total bolus administration and the effective number of PCIA bolus administrations in the first 6 postoperative hours were much lower in the EA group than in the control group (Table 2), whereas there were no differences 6–48 hours postoperatively. From the first 6 hours to the second postoperative day, there were no significant differences in the mean total bolus administration and the effective number of bolus administrations between the 2 groups. There was no difference in the total fentanyl dose between the EA and control groups (0.67 ± 0.09 mg vs. 0.73 ± 0.12 mg, respectively).

Table 2. PCIA Bolus Administration in the Two Groups (Mean ± SD)

	Group A ($n = 41$) (Electro-Acupuncture Group)	Group C ($n = 40$) (Control Group)
Total PCIA Bolus Administration		
0–6 h	14.43 ± 10.67*	24.87 ± 14.88
6–48 h	8.30 ± 2.91	9.63 ± 3.21
Effective Number of PCIA Bolus Administrations		
0–6 h	4.83 ± 2.95*	9.67 ± 3.35
6–48 h	5.64 ± 1.23	5.57 ± 1.52

Notes: Data are expressed as mean ± SD or numbers. *$p < 0.05$ compared to the control group. PCIA = patient controlled intravenous analgesia.

Secondary Parameters

On the first postoperative day, the incidence of PONV was 14.6% and 32.5% in the EA and control groups, respectively, with no significant difference between the groups ($\chi^2 = 3.599$, $p = 0.058$). On the second postoperative day, the rate of PONV greatly decreased to 4.9% and 10.0% in the EA and control groups, respectively ($\chi^2 = 0.744$, $p = 0.379$).

Table 3 contains the incidence and degree of dizziness and feeling of fullness in the head. During the first 6 postoperative hours, most patients experienced a severe to moderate feeling of dizziness and fullness in the head, and this feeling were more serious in the control group than in the EA group ($p < 0.05$). Within 6–24 postoperative hours, more patients in the control group suffered from moderate to severe dizziness than in the EA group ($p = 0.001$). However, these uncomfortable feelings were largely resolved by the next day in both groups.

Almost all patients could not drink water within 6 hours post-surgery and were given only very little water for lips and mouth humidification. We found that the patients in the EA group could eat liquid foods much sooner than those in the control group, which indicates that they had a better appetite (Fig. 2). In particular, during the first 24

Table 3. Dizziness and Feeling of Fullness in the Head (Cases/%)

		None		Mild		Moderate		Severe	
	Groups	Cases	Ratio (in %)	Cases	Ratio (in %)	Cases	Ratio (in %)	Cases	Ratio (in %)
0–6 h after operation	Group A	5	12.2	11	26.8	15	36.6	10	24.4
	Group C	2	5	4	10	19	47.5	15	37.5
6–24 h after operation	Group A	10	24.4	16	39.0	13	31.7	2	4.9
	Group C	2	5.0	10	25	20	50	8	20.0
24–48 h after operation	Group A	27	65.9	9	22.0	4	9.8	1	2.4
	Group C	20	50.0	7	17.5	9	27.5	2	5.0

Notes: Data are expressed as numbers and percentages. During 0–6 hrs and 6–24 hrs after operation, the difference between two groups was significant, $p < 0.05$. Group A = Electro-acupuncture group, Group C = control group.

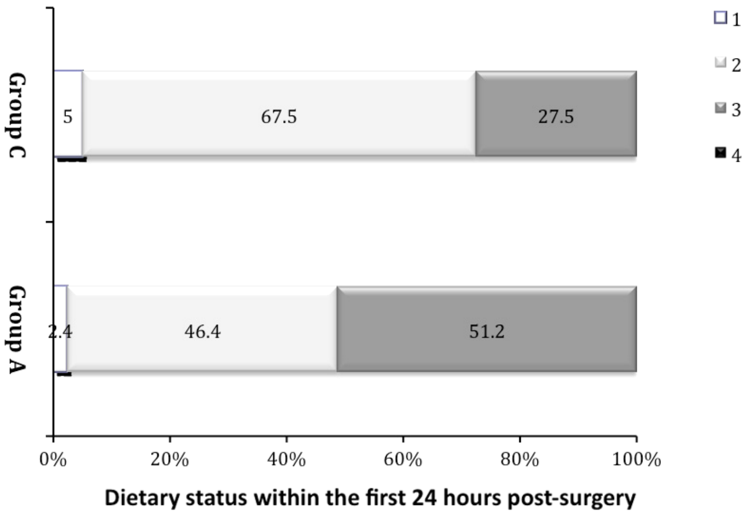

Figure 2. Dietary status within the first 24 hours post-surgery. In group A, more patients were able to consume a sufficient liquid diet than in group C (the ratio was 51.2% vs. 27.5%). Using the non-parametric test and the rank sum test, $p = 0.03$. 1 = unable to eat, 2 = very few liquid foods or water, 3 = sufficient liquid foods or some solid foods and 4 = normal eating. Group A = electro-acupuncture group, Group C = control group.

postoperative hours, significantly more patients in the EA group than in the control group were able to eat a sufficient liquid diet (51.2% vs. 27.5%) ($p = 0.03$). In both groups, most patients began to eat solid foods, such as eggs, on the third morning. More patients exhibited mild or moderate anorexia in the control group than in the EA group; however, this difference was not statistically significant ($p = 0.058$) (Fig. 3).

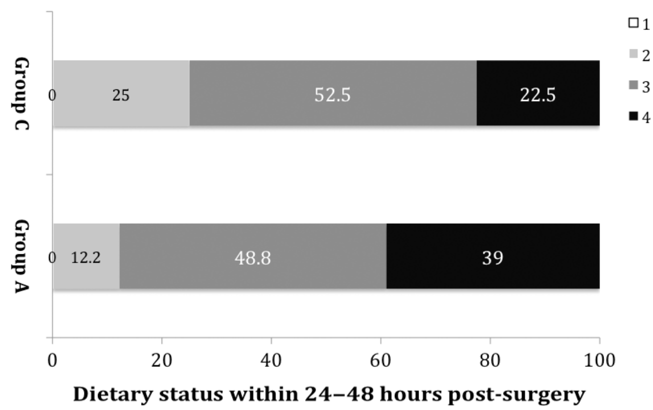

Figure 3. Dietary status 24–48 hours post-surgery. In group A, more patients were able to consume a sufficient liquid diet and even eat a normal diet than in group C (39% vs. 22.8%). Using a non-parametric test and the rank sum test, $p = 0.058$, with no differences between the groups. 1 = unable to eat, 2 = very few liquid foods or water, 3 = sufficient liquid foods or some solid foods and 4 = normal eating. Group A = electro-acupuncture group, Group C = control group.

Discussion

The main finding of the present study is that EA reduces postoperative pain and decreases the number of effective PCIA bolus administrations when used as an adjuvant to general anesthesia during a supratentorial tumor resection. Moreover, EA also improves appetite recovery and decreases some uncomfortable side effects, such as dizziness and feeling of fullness in the head.

Based on the VAS score, we found that the postoperative pain of neurosurgery patients was, on average, greatest on the day of surgery, especially within 24 hours. This study revealed that the pain after neurosurgery was moderate and decreased over time. This finding is similar to other studies that have demonstrated a reduction in postoperative pain over time (Sommer *et al.*, 2008; Yeh *et al.*, 2010; Dong *et al.*, 2013). The use of EA during anesthesia contributes to the achievement of 2 important aims. First, it reduces the effective number of PCIA bolus administrations and reduces the dose of conventional drugs (e.g., fentanyl) and their side effects. Second, EA helps to maintain pain intensity within a tolerable range. The differences in the VAS score and number of PCIA bolus administrations between the 2 groups were significant, especially in the first 6 hours after the operation. This result is somewhat different from other similar studies, which have demonstrated that EA continuously improves acute postoperative pain up to the third postoperative day (Yeh *et al.*, 2010). The reasons for this difference may include differences in the surgical site and the frequency and duration of the EA application (Lin and Chen, 2008; Leung, 2012). Both clinical and laboratory results indicate, after the EA stimulation, modulation of the noradrenaline and 5-HT signaling system, the production of endogenous neuropeptides acting upon the μ-opioid and N/OPQ receptors, and the production of somatostatin and other related neurotrophins that together enhance the descending inhibition of nociception on the spinal afferents (Leung, 2012). But there was no definitive evidence regarding the duration of this analgesia effect of EA after the discontinuation of stimulation. In the present study, the supplementary analgesic effect of EA was concentrated within the first 6 postoperative hours, but the other effects, such as improving recovery, were maintained for a longer time.

The choice of acupoints in our study was based on the principle of "acupoints along meridians" (Han, 1997) from the theory of "treating disease of the upper part by managing the lower". In our previous studies, we found that EA combined with general anesthesia decreases the anesthetic requirement and hastens recovery after anesthesia. In addition, the most effective acupoints used in supratentorial neurosurgery are LI4 and SJ5, BL63 and LR3, and ST36 and GB40 (An *et al.*, 2010a,b; Yang *et al.*, 2012). Therefore, we applied the same acupoints and stimulation intensity in the present study to obtain the best adjunct analgesia effect for postoperative pain.

Many previous studies have demonstrated that EA may decrease the incidence of PONV. Wang *et al.* found that EA stimulation at the P6 points is an effective adjunct to anti-emetic drug therapy for PONV in patients undergoing supratentorial craniotomy (Wang *et al.*, 2010), and Lv also obtained the same result (Lv *et al.*, 2013). EA stimulation may mediate the release of bendorphins in the cerebrospinal fluid, activate serotonergic and norepinephrinergic fibers, and change the serotonin transmission. Enhancing gastric motility is also a possible role for central dopaminergic receptors in EA's anti-emetic function (Wang *et al.*, 2010). Although in the present study, we found that the patients who received EA stimulation had a lower incidence of PONV compared to the control group (14.6% vs. 32.5%), the difference was not significant ($p = 0.058$). This result may be explained by the preventive use of the anti-emetic drug ondansetron in the PCIA, which may lead to a low incidence of PONV.

Another primary clinical advantage of EA is the improved postoperative recovery, including a reduction in the degree of uncomfortable side effects, such as dizziness and feeling of fullness in the head, and improved appetite restoration. In this study, EA decreased the degree of dizziness within 24 hours postoperatively. In fact, these unpleasant side effects are very common and even serious in neurosurgery patients. There are 2 possible explanations for this observation. First, EA reduces the postoperative pain that may produce these unpleasant side effects; second, the conditioning effects of traditional Chinese acupuncture through its regulation of inherent neuro–endocrine–immune (NEI) network, are possible explanations for this phenomenon (Ding *et al.*, 2014). EA stimulation has a long history of improving appetite (Yang *et al.*, 2013). In this study, after EA stimulation, more patients were able to eat a sufficient liquid diet than patients in the control group (51.2% vs. 27.5%). Undoubtedly, this will promote patient recovery.

A very important advantage of the present study is that it is double-blinded. In fact, most Chinese studies regarding acupuncture during an operation were not truly double-blinded, which may affect the objectivity and accuracy of the results. In this study, stimulation was applied after the patients were sedated; therefore, they were not aware of their EA status. In addition, the physician who observed the pain status and other operative outcomes was also unaware of the patients' EA status. To avoid bias, all the data were collected by the same person asking the same questions. However, the results are limited by the fact that the data obtained relied on patient reporting, which may be affected by the patient's level of education and understanding.

In summary, the use of EA in supratentorial tumor resection patients significantly reduces postoperative pain and decreases the effective number of PCIA bolus administrations. Moreover, EA also improves appetite recovery and decreases the occurrence of uncomfortable side effects, such as dizziness and feeling of fullness in the head.

Acknowledgment

Financial support for this study was provided by the National Basic Research Program of China (973 Program), No. 2007CB512503.

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